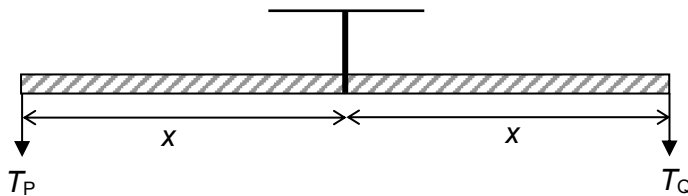


magnitude of the force on the floor by the boy.

5. Ans: B

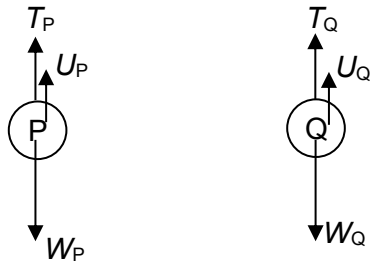


At equilibrium,

$$CW = ACW$$

$$T_P x = T_Q x$$

$$T_P = T_Q$$



$$T_P = T_Q$$

$$W_P - U_P = W_Q - U_Q$$

$$W_P - \rho_X gV = W_Q - \rho_Y gV$$

Since $\rho_X > \rho_Y$, $U_P > U_Q$ since both has the same volume.

Therefore, $W_P > W_Q$, $m_P > m_Q$.

